8-1-1: Examining SDN and SD-WAN

At the end of this episode, I will be able to:

1. Identify and explain the benefits of SDN and SD-WAN technologies, given a scenario.

Learner Objective: *Identify and explain the benefits of SDN and SD-WAN technologies, given a scenario*

Description: In this episode, the learner will examine Software-Defined Networks, or SDNs, and Software-Defined Wide Area Networks, or SD-WAN. We will explore the benefits and components of these evolving network technologies.

- What is SDN?
    - Software-Defined Network (SDN)
        - Decouples network control and forwarding functions for more flexible management.
        - Enables programmatically efficient network configuration and management.
- What are the SDN Layers:
    - Management plane
        - Engages directly with network applications.
    - Control plane
        - Hosts the SDN controller, orchestrating flow control and policy application.
    - Data plane
        - Consists of the physical and virtual network devices.
- What are the SDN Controllers:
    - The central authority that manages traffic flow and policy enforcement across the network.
    - Can be a physical or virtualized device
- What are the NB and SB interfaces?
    - Northbound interface (NBI) - Connects the SDN controller with the applications; utilizes RESTful APIs for communication, enabling developers to program the network using high-level languages like Python and JavaScript.
    - Southbound interface (SBI) - Links the SDN controller with the network devices; OpenFlow is a predominant protocol here, facilitating the controller's ability to direct network traffic.
- What are OpenFlow and RESTful API?
    - OpenFlow - A foundational protocol for SDN, enabling direct access and manipulation of the forwarding plane of network devices.
    - RESTful APIs - Architectural style used for designing networked applications, allowing for easy and dynamic interactions between the SDN controller and applications via HTTP requests.
- What is SD-WAN (Software-Defined Wide Area Network)
    - Extends SDN concepts to wide area networks, overlay network over large geographical areas.
    - Focused on delivering a high-performance WAN that uses multiple transport technologies.
- What are some of the SD-WAN components?
    - Overlay Network - Virtualizes WAN connections to connect different parts of the network over the internet or a private network.
    - Edge Devices - Sit at the perimeter of a network, providing connectivity and SD-WAN functionalities.
    - Controllers:
        - SD-WAN Controller - Oversees traffic routing, policy management, and the health of the SD-WAN connections.
    - Interfaces:
        - Management Interface - For SD-WAN configuration and monitoring.
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Orchestration Interface - Manages and synchronizes policies and settings across the SD-WAN fabric.

- Additional Resources:
    - If applicable